Assessing the Impact of Project ECHO: A Distributional Semantics Approach to Rural Healthcare Training Evaluation

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Intro/Project Purpose

Project ECHO (Extension for Community Healthcare Outcomes) is a program that started in 2003 in New Mexico and is dedicated to empowering rural health providers to effectively manage complex patients by training providers serving these communities. **When health providers are empowered to treat patients locally that they originally would have referred to a specialty care center:**

- Travel to a different care center is eliminated, speeding up treatment
- Large care centers like the University of Utah receive less referrals from rural areas, allowing them to improve quality of care and avoid being overwhelmed
- Care providers are more likely to be familiar with a patient's history, which can lead to improved care

The University of Utah was the third place in the world to replicate this model in 2011, and there are now 3,000 programs across the world. It has become important to assess the impacts of these trainings by measuring health outcomes within these communities. The current study explores the use of distributional semantic approaches for multi-level assessment of the Utah program's effectiveness, bridging the gap between healthcare training and patient outcomes [1].

Discussion

By linking Project ECHO didactic titles to health indicators through concept overlap using the Concept Bag method, we developed a novel method for measuring the impact of healthcare training initiatives over time using publicly available data [2]. This approach of semantic matching provides a means to integrate novel data resources, as well as assessing the impacts of multi-level interventions in absences of planned data collections for program evaluations.

However, because publicly available data sources tend to be high-level, this technique would be greatly improved through partnerships with clinics to produce more granular data.

Conclusions

1. Novel data sources can be reliably tied to training topics using concept bagging, which is useful in the absence of direct data collection.
2. Linking training presentation titles to health indicators can be done computationally, making the process more objective and effective
3. In the absence of a data collection strategy, Project ECHO needs to focus more trainings on topics that are measured at the state level to help evaluate overall program effectiveness

References

[1] Cohen, T., & Widdows, D. (2009). Empirical Distributional Semantics: Methods and Biomedical Applications. Journal of Biomedical Informatics, 42(2), 390. https://doi.org/10.1016/J.JBI.2009.02.002
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[3] Aronson, A. R., & Lang, F. M. (2010). An overview of MetaMap: Historical perspective and recent advances. JAMIA, 17(3), 229–236. https://doi.org/10.1136/jamia.2009.002733

Linking of Didactic Titles to Indicator

We calculated the number of both didactic titles and health indicator titles that had a Key Concept. By creating a list of useful concepts present in both sets we were able to calculate meaningful overlap. An example of a link between a didactic title and a health indicator is shown below.

The function of this step is three-fold. Project ECHO can investigate if a particular training seems to have had an effect on state data by using a link, they can determine how well their trainings match what the state has decided is important to track, and they can create new trainings based on what will be most measurable.

Program	Titles linked to data	% of total
Pregnancy	92	56%
Behavior	17	20%
Hep C	16	18%
GI/Liver	17	17%
Pediatrics/Telehealth	24	18%
Total/Average	166	26%
Data Source	**Indicators linked to title**	**% of total**
IBIS	48	24%
UDOHHS	1	25%
KFF	79	41%
CDC NCH	4	22%
Total/Average	143	28%

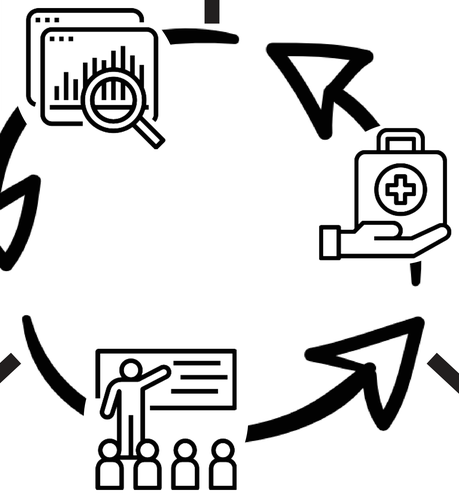
Linkage rates between heath indicators and didactic titles. About a quarter of all didactic titles could be linked to a publicly available health indicator, and vice versa.

Example Link

Title: Lowering the Cesarean Delivery Rate

Key Concept: Cesarean section

Health Indicator: Cesarean delivery among low risk women with no prior births



Public Health Indicators

Data Source	Indicator Titles	Concepts	Unique Concepts	Key Concepts
IBIS	197	692	423	55
UDOHHS	4	4	4	1
KFF	194	1213	340	81
CDC NCH	18	74	31	6
Total	413	1983	798	143

Indicator Title Medical Concept Extraction. The number of key concepts that were present in the publicly available health indicator data shows the extent to which Project ECHO didactics are addressing key health measures.

After extracting medical concepts from the didactic titles we performed a similar analysis on public health data sources, including Utah's Public Health Indicator Based Information System (IBIS), Utah Department of Health & Human Services (UDOHHS) Data & Reports, Kaiser Family Foundation (KFF) State Reports, and the Center for Disease Control and Prevention's National Center for Health Statistics (CDC NCH).

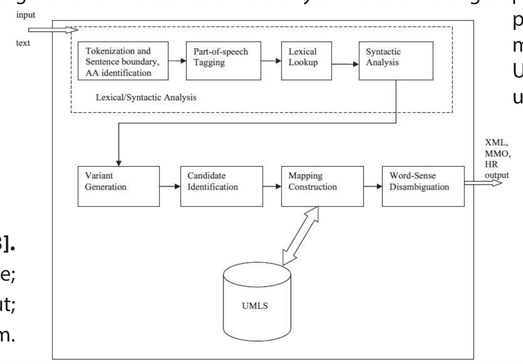
The titles for indicators shown on these sites by graphs or reports were used as input text for MetaMap. We determined ***Key Concepts*** (shown right) by manually filtering concepts present in indicator titles and didactic titles for specificity, ie. "Child" is too general to be a Key Concept, but "Breast Feeding" is specific and can be influenced by training.

Key Concepts

- Acute hepatitis C
- Aniridia type 1
- Asthma
- Attention deficit hyperactivity disorder
- Breast Feeding
- COVID-19
- Cerebrovascular accident
- Cesarean section
- Childbirth
- Chronic liver disease
- Colorectal Carcinoma
- Congenital Abnormality
- Deafness
- Dependence
- Depressive disorder
- Disability
- Exercise
- Family history: Hypertension
- Feeling suicidal (finding)
- Fetus
- Gestational Diabetes
- HIV
- HIV Infections
- Hearing Loss, Partial
- Hepatitis A
- Hospitalization
- Hypertensive disease
- Influenza
- Influenza virus vaccine
- Liver Cirrhosis
- Malignant Neoplasms
- Marihuana
- Maternal Mortality
- Mental Depression
- Mental disorders
- Obesity
- Obstetric Delivery
- Opioid Use Disorder
- Perinatal period
- Pharmaceutical Preparations
- Pharmacologic Substance
- Pregnancy
- Premature Birth
- Prenatal care
- Sex Behavior
- Sexually Transmitted Diseases
- Sleep
- Smoking in Pregnancy
- Syphilis
- Traumatic injury

Treatment Improvement Didactics

We initiated our investigation by analyzing 571 Project ECHO didactic titles across five focus areas: Pregnancy, Behavior, Hepatitis C, Gastroenterology and Liver, and Pediatrics/Telehealth. All titles for all didactics given since the program's inception were included, so there is redundancy where the same didactic was given over the course of several years or to several groups. Employing the NIH's MetaMap natural langauge processing tool, we extracted named entities and mapped medical concepts from the titles to concepts within the Unified Medical Language System (UMLS), yielding 1050 unique concepts.



MetaMap system diagram [3].
HR, human readable;
MMO, MetaMap machine output;
UMLS, unified medical language system.

Program	Didactics	Concepts	Unique Concepts
Pregnancy	164	559	353
Behavior	86	248	172
Hep C	87	347	199
GI/Liver	101	362	230
Pediatrics/Telehealth	133	582	367
Total	571	2098	1050

Didactic Title Medical Concept Extraction.
Programs were the five areas of focus for Project ECHO in Utah. A "concept" can be linked to several words. ie. "Mental disorders" can be linked to "anxiety", "depression", etc.